

Gabe Flanagin

Bellingham, WA
flanagg@wwu.edu - (206)743-
1233

Education

Western Washington University, Bellingham, WA

M.S. in Mathematics

September 2025 - Present

Western Washington University, Bellingham, WA

B.S. in Mathematics, Minor in Chemistry

September 2021 - June 2025

Research Interests

Applied and computational mathematics; optimal control; numerical methods for PDEs; scientific computing; computational materials science; energy-related modeling

Publications

Flanagin, G.; Berger, R.F. *Rationalizing Patterns in the Cation Ordering, Geometric Distortion, and Electronic Properties of a Class of I-V-VI₂ Chalcogenide Semiconductors*.

Journal of Physical Chemistry C, 2026, available online. <https://doi.org/10.1021/acs.jpcc.5c05853>

Research Experience

Research Assistant - Berger Research Group

WWU Department of Chemistry

June 2024 - June 2025

- Performed first-principles electronic structure calculations using density functional theory (DFT) to study chalcogenide semiconductors relevant to photovoltaic applications
- Conducted structural relaxation, stability analysis, and electronic property characterization (band structures, density of states, electron localization functions)
- Designed and automated computational workflows using Bash scripting
- Analyzed and visualized simulation data using Python
- Ran large-scale calculations on a university high-performance computing cluster
- Collaborated closely with PI through weekly research meetings, resulting in a peer-reviewed publication

Senior Project - A Max-Plus Algorithm for Optimal Control of Dynamic Systems

WWU Department of Mathematics

January 2025 - June 2025

- Studied a max-plus algebra based algorithm for approximating the solution to high-dimensional Hamilton-Jacobi-Bellman equations
- Implemented the algorithm in Python (NumPy, SciPy, Matplotlib)
- Proved a generalization of the algorithm by weakening a semi-convexity assumption required for convergence.
- Authored a 25-page technical report and delivered a 20-minute presentation at WWU Scholars' Week

Presentations

- *An Introduction to Spatiotemporal Rabies Models*, WWU Mathematics Colloquium Series, 2026
- *Using Max-Plus Algebra to Compute Reachability in Dynamic Systems*, WWU Scholars' Week, 2025
- *Rationalizing Patterns in the Cation Ordering, Geometric Distortion, and Electronic Properties of a Class of I–V–VI₂ Chalcogenide Semiconductors*, WWU Scholars' Week poster presentation, 2025

Programming Languages & Tools

- Python (NumPy, SciPy, Matplotlib; scientific computing & numerical methods)
- MATLAB
- Bash scripting (workflow automation)
- VASP (electronic structure simulations)
- Experience with high-performance computing environments

Teaching & Tutoring Experience

Graduate Teaching Assistant

WWU Department of Mathematics

September 2025 - Present

- Teach undergraduate courses in algebra and pre-calculus

Lead Math Tutor

WWU Libraries Tutoring Center

April 2024 - June 2025

- Facilitated weekly math review sessions for peer tutors
- Trained new tutors in group facilitation and effective pedagogical strategies
- Served as liaison between tutors, staff, and faculty to improve student support

Peer Tutor

WWU Libraries Tutoring Center

September 2022 - April 2024

- Tutored mathematics, statistics, chemistry, and physics in a drop-in setting
- Led weekly study groups for introductory math and chemistry courses

Tutor

Hawthorne Learning Solutions

November 2022 - June 2024

- Provided one-on-one tutoring in high school mathematics and science
- Emphasized conceptual understanding, study strategies, and problem-solving skills